Asymmetric Process-Reward Adaptation for Mathematical Reasoning

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Abstract

Correct answers can conceal faulty derivations. We propose asymmetric process reward model (PRM) adaptation: retain failure supervision and public step labels while omitting positive outcome supervision. Four matched GRPO experiments test whether this improves mathematical reasoning.

1 Motivation and research question

Mathematical RL needs reliable rewards to favor valid reasoning over lucky answers. Yet answer-correct solutions can contain errors (Zheng et al., 2025), so outcome labels may certify flawed derivations. We ask: **can withholding positive outcome supervision improve downstream policy accuracy?** Reduced false certification must outweigh lost supervision. A useful rule would adapt public PRMs without new annotations; diagnostics characterize failures without uniquely attributing them to label noise or policy drift.

2 Related work and contribution

PRM800K provides costly human step labels on previously sampled solutions (Lightman et al., 2023); their coverage need not track a changing policy. PRIME updates implicit rewards online from outcome labels (Cui et al., 2025), which do not identify valid individual steps. PURE's min-form credit assignment mitigates but does not eliminate PRM reward hacking (Cheng et al., 2025). LCA uses softmax-weighted pooling and tests controlled answer-correct/process-wrong label noise (Jia et al., 2026, App. E.3). Our contribution is **withholding weak positive labels while retaining step supervision**, tested through downstream GRPO. C adopts LCA-inspired pooling, not its full method; D–C isolates the removed loss term.

3 Proposed method

Offline adaptation. Sample four solutions on each of 512 training problems from the initial policy; retain complete, unambiguously verified trajectories. For native step-boundary correctness scores $p_{\phi,t}$, set $e_{\phi,t} = 1 - p_{\phi,t}$. Following LCA,

$$w_t = \frac{\exp(e_{\phi,t}/T)}{\sum_j \exp(e_{\phi,j}/T)}, \qquad E_\phi(\tau) = \sum_t \mathrm{sg}(w_t) e_{\phi,t}, \qquad T = 0.2. \tag{1}$$

Here sg stops gradients through weights. Define $L_{\text{fail}} = \mathbb{E}_{\tau^-}[-\log E_\phi(\tau^-)]$ and $L_{\text{success}} = \mathbb{E}_{\tau^+}[-\log(1 - E_\phi(\tau^+))]$, taking separate means over failures/successes. A shared set of 2,048 public positive/negative step labels supplies mean binary cross-entropy L_{step}:

$$L_c = L_{\text{fail}} + cL_{\text{success}} + L_{\text{step}}, \qquad c_C = 1, \quad \boxed{c_D = 0 \text{ (ours)}}. \tag{2}$$

Failure labels weakly indicate an error without locating it. Step labels discourage blanket rejection but cannot guarantee it is avoided.

Preprint.

Policy training. Freeze each PRM throughout RL. For complete, parseable responses in B–D, compute

$$R_i = y_i + 0.2 \min_t p_{\phi,t}(\tau_i), \qquad \hat{A}_i = \frac{R_i - \bar{R}}{\sigma_R + 10^{-4}}. \tag{3}$$

Here y_i is binary correctness; $\bar{R}, \sigma_R$ denote the group mean/sample standard deviation. A uses $R_i = y_i$; invalid/truncated outputs get zero in all arms. GRPO applies detached advantages to completion tokens (Shao et al., 2024). For valid same-outcome groups, $\hat{A}_i = (m_i - \bar{m})/(\sigma_m + 5\times10^{-4})$, where $m_i = \min_t p_{\phi,t}$: 0.2 does **not** bound normalized PRM influence. Unlike A, B–D can reward complete wrong answers over invalid outputs; C/D share this rule.

4 Experimental design and expected findings

Arm	Reward and PRM preparation	Question tested
A	Outcome only; no PRM	Reference GRPO performance
B	Outcome + original frozen PRM	B–A: value of an existing PRM
C	Outcome + PRM adapted with $c = 1$	C–B: complete adaptation package
D (ours)	Outcome + PRM adapted with $c = 0$	**D–C: value of the proposed change**

Models and controls. Use Qwen2.5-Math-1.5B (policy LoRA rank 16), the H4 1.5B PRM (PRM LoRA rank 8) (Hugging Face H4, 2025), and TRL 0.23.1/PEFT GRPO with group scaling (Hugging Face, 2026). Outcome-only execution is verified on one 80 GB A100. A **PRM-only control S** uses L_{step} with matched initialization, anchor batches and updates: S–B tests anchor adaptation; D–S tests failure supervision. No RL arm is added.

Pre-RL check. Score shared initial-policy rollouts on 64 validation questions (eight each) with B/C/D/S; audit missing updates. Before B–D RL, fix replication using C–D advantages: if sign agreement is at least 99% and every group's RMS difference is at most 0.05, use seed 17 only across A–D; otherwise use 17/42/83. This is a budget heuristic, not evidence of equivalence; test accuracy and ProcessBench never choose replication.

Data. Use MATH (6,993 train/500 validation) (Hendrycks et al., 2021), MATH500 and GSM8K tests (Cobbe et al., 2021). Both tests have zero exact normalized overlap with the public PRM800K inventory. Full MATH overlaps PRM800K (OpenAI, 2023) and is supplementary. Exclude validation/test/diagnostic questions from new anchors; unknown inherited exposure and paraphrases remain limitations.

Budget and matching. C/D share PRM initialization, batches, optimizer and updates; only c changes, without loss re-averaging. All arms share policy initialization, prompt order and decoding: eight rollouts, temperature 1.0, 2,048-token cap. Fix **24M completion tokens/run**, with 6M/12M/24M milestones: 96M or 288M total under the gate. Count failures/truncations; report updates, prompts, lengths and GPU hours including PRM/precheck costs.

Evaluation and power. Use identical greedy decoding without reranking. Primary: **D–C MATH500 accuracy at 24M**; D–A/B tests practical value. Report paired-seed differences and 95% question-cluster bootstrap intervals retaining all seeds together; mean/std only with three seeds. Single-seed results are exploratory. At an assumed 10% disagreement rate, the approximate single-seed 80%-power detectable difference is 4 percentage points (two-sided $\alpha = .05$). Small effects may remain unresolved; milestones do not select the endpoint.

Mechanism and risks. ProcessBench MATH diagnostics include first-error accuracy, correct-chain acceptance, their harmonic mean at threshold 0.5, and valid-chain AUROC among answer-correct solutions (406 valid/94 flawed). Report paired question-bootstrap intervals for metrics/differences and overlap strata; small uncertain differences remain inconclusive. On shared rollouts, report D–C rewards/advantages within fixed all-correct/all-wrong/mixed groups. Online group rates are descriptive, not causal subgroup effects. Flag invalid outputs; monitor steps and missing delimiters with a fixed double-newline splitter. Diagnostics never tune PRMs or establish policy gains alone.

Tentative timeline. Week 1: buffer, anchors, PRM controls and precheck. Weeks 2–3: C/D first, then remaining A/B for seed 17; add paired seeds only as predeclared. Week 4: diagnostics, statistics and report; disclose unfinished runs.

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